2023 SSMO Speed Round

SMO Team

SR 1: Let $S_1 = \{2, 0, 3\}$ and $S_2 = \{2, 20, 202, 2023\}$. Find the last digit of

$$\sum_{a \in S_1, b \in S_2} a^b.$$

SR 2: Points A, B, C are chosen independently and uniformly at random inside the square with vertices $(-1, -1)$, $(-1, 1)$, $(1, -1)$, and $(1, 1)$. The probability that the centroid of triangle ABC lies in the first quadrant can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

SR 3: Pigs like to eat carrots. Suppose a pig chooses 6 letters one at a time, independently and uniformly at random from the set $\{c, a, r, o, t\}$, to form a "word". If the 6 letters do not spell "carrot", the pig gets frustrated and tries to spell it again (by rechoosing all 6 letters). What is the expected number of tries it takes for the pig to spell "carrot"?

SR 4: Let $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for all $n \geq 3$ be the Fibonacci numbers. If distinct positive integers $a_1, a_2, \ldots, a_m$ satisfy $F_{a_1} + F_{a_2} + \cdots + F_{a_m} = 2023$, find the minimum possible value of $a_1 + a_2 + \cdots + a_m$.

SR 5: In parallelogram $ABCD$ with $AB = 6$ and $BC = 8$, a point P inside the parallelogram is chosen such that $\angle APD + \angle BPC = 180^\circ$. Find the sum of the maximum and minimum values of $(PA)(PC) + (PB)(PD)$.

SR 6: Find the smallest odd prime that does not divide $2^{75!} - 1$.

SR 7: At FenZhu High School, 7th graders have a 60% chance of having a dog and 8th graders have a 40% chance of having a dog, independently of each other. Suppose there is a classroom of 30 7th graders and 10 8th graders. If exactly one person owns a dog, then the probability that a 7th grader owns the dog can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

SR 8: Circle ω has chord AB of length 18. Point X lies on chord AB such that $AX = 4$. Circle ω_1 with radius r_1 and ω_2 with radius r_2 lie on two different sides of AB. Both ω_1 and ω_2 are tangent to AB at X and internally tangent to ω. The sum of the maximum and minimum values of $r_1 r_2$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

SR 9: Find the sum of the maximum and minimum values of $8x^2 + 7xy + 5y^2$ under the constraint that $3x^2 + 5xy + 3y^2 = 88$.

SR 10: In a circle centered at O with radius 7, we have non-intersecting chords AB and DC. O is outside of quadrilateral $ABCD$ and $AB < CD$. Let X be the intersection of lines AO and CD, and let Y be the intersection of lines BO and CD. Suppose that $XO + YO = 7$. If $YC - DX = 2$ and $XY = 3$, then AB can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.